

Author: It is a good idea. Let us analyze, for the sake of convenience, the graph of the function $y = \sqrt{x}$ (Fig 25). This figure illustrates only two situations. One of them represents the selection of ε_1 (see the figure). It is easy to infer that δ_1 is the value that we look for: the values of the function at all points x from the δ_1 -neighbourhood of point a are inside the interval $]b - \varepsilon_1, b + \varepsilon_1[$. These values are represented by the portion of the graph between points A and B .

The second situation represents the selection of ε_2 . In this case the number that we seek for is δ_2 : the values of the function at points x from the δ_2 -neighbourhood of point a are represented by the portion of the graph between points A' and B' .

Reader: Everything you have just described looks so obvious that I see no "cream", to use your own words.

Author: "The cream" consists in the following. No matter how small $]b - \varepsilon, b + \varepsilon[$ is, one may always select a δ -neighbourhood for point a such that for all points x in this δ -neighbourhood (all points, with the exception of point a itself and those at which the function is not defined) the values of the function should by all means lie within the indicated interval. V. Imp.

Reader: Could you give an example of a function violating this rule?

Author: For instance, the function $y = \sin \frac{1}{x}$ in the vicinity of point $x = 0$. The graph of the function is plotted in Fig. 2A. Obviously, the smaller is $|x|$ the greater is the frequency with which the graph of the function oscillates about the x -axis. For an infinitely small $|x|$ the frequency of the oscillations tends to infinity. It is easy to prove that the function $y = \sin \frac{1}{x}$ has no limit at $x = 0$.

Reader: But this function is not defined at zero.

Author: You are right. However, this fact is irrelevant from the viewpoint of the existence (or absence) of the limit of the function at $x = 0$. This function is defined over $]-\infty, 0[$ and $]0, \infty[$. Point $x = 0$ is a common boundary between the intervals over which the function $\sin \frac{1}{x}$ is defined.

V. Imp. Please remember it